SDA TAP Lab Project:
Processing Uncorrelated Tracks (UCTs) in the Common Task Framework: Labelling & Data Storage

SDA TAP Lab
Space Systems Command
United States Space Force

SPACE DOMAIN AWARENESS

Tools, Applications, & Processing Lab

SDA TAP Lab Overview

Space Domain Awareness (SDA) - Rapidly predict, detect, track, identify, warn, characterize, and attribute, threats to U.S., commercial, allied, and partner space systems.

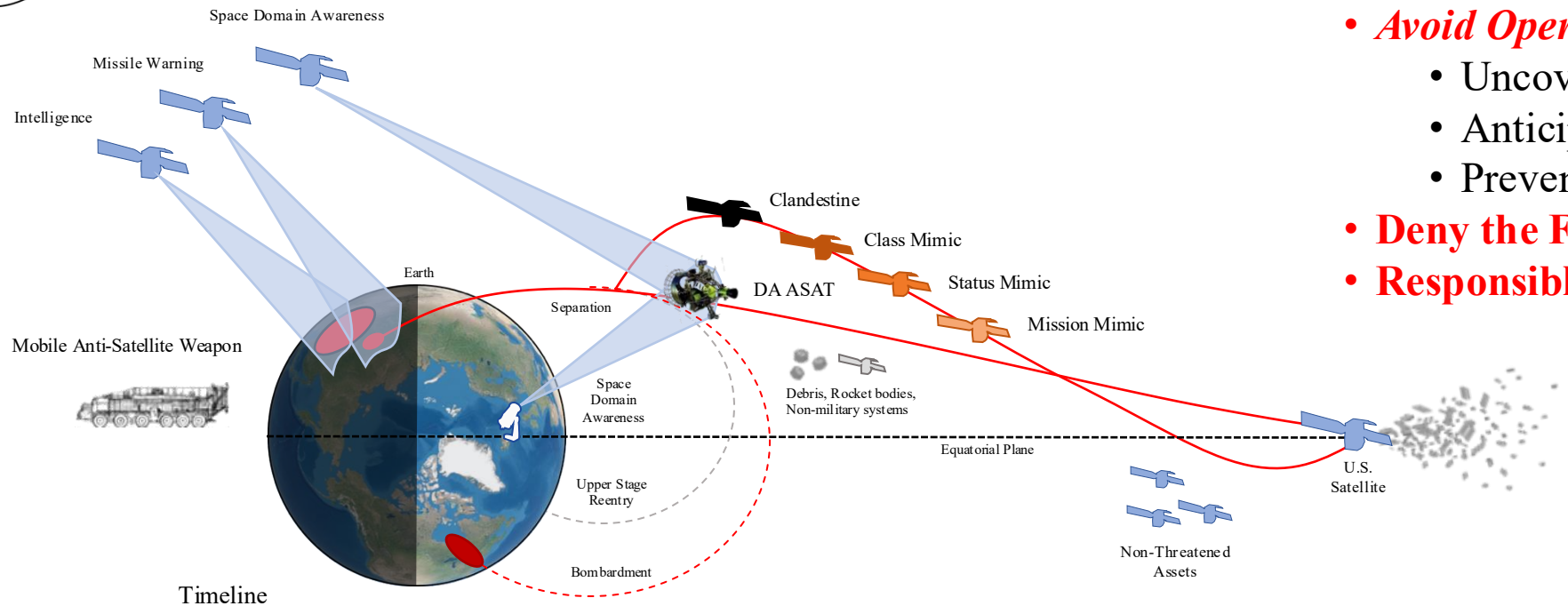
Mission: SDA TAP Lab accelerates the delivery of space battle management software to operational units. We decompose kill chains, prioritize needs with operators, map needs to technologies, and onboard tech to existing platforms quickly. We partner with industry, academia, and across the government to succeed.

A scalable way to onboard, deploy, sustain "apps to close gaps"

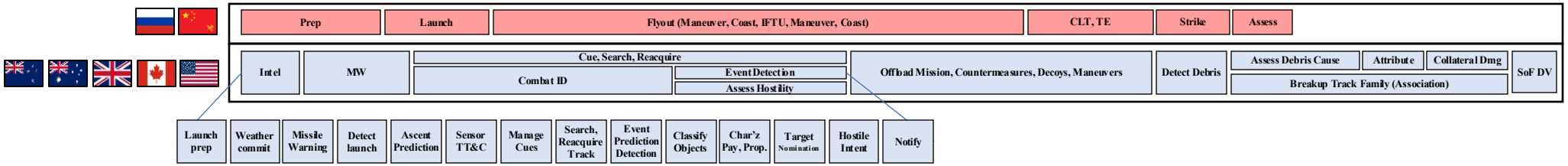
SDA TAP Lab

Theory of Success: Competitive Endurance

- ***Avoid Operational Surprise***
  - Uncover surprise vectors
  - Anticipate, deter, mitigate hostility
  - Prevent unnecessary escalation
- **Deny the First-mover Advantage**
- **Responsible Counterspace Campaigning**



Timeline



Gen Saltzman
CSO

Gen Guetlein
VCSO

Col Agrawal
Commander

Barb Golf
HQE

Surprise comes via camouflage, concealment, deception, and maneuver (CCDM). **We must interrogate targets for evidence of CCDM.**

Apollo Accelerators

- 3-month Apollo Accelerators
- Decompose Kill Chains
- Industry, Academia, Government
- Multi-vendor integrated solution
- Deploy to NSDC and JCO
- Transition to Programs of Record

Apollo Accelerator Cohort 2 is dividing parts of the kill chain amongst team members

Project Background

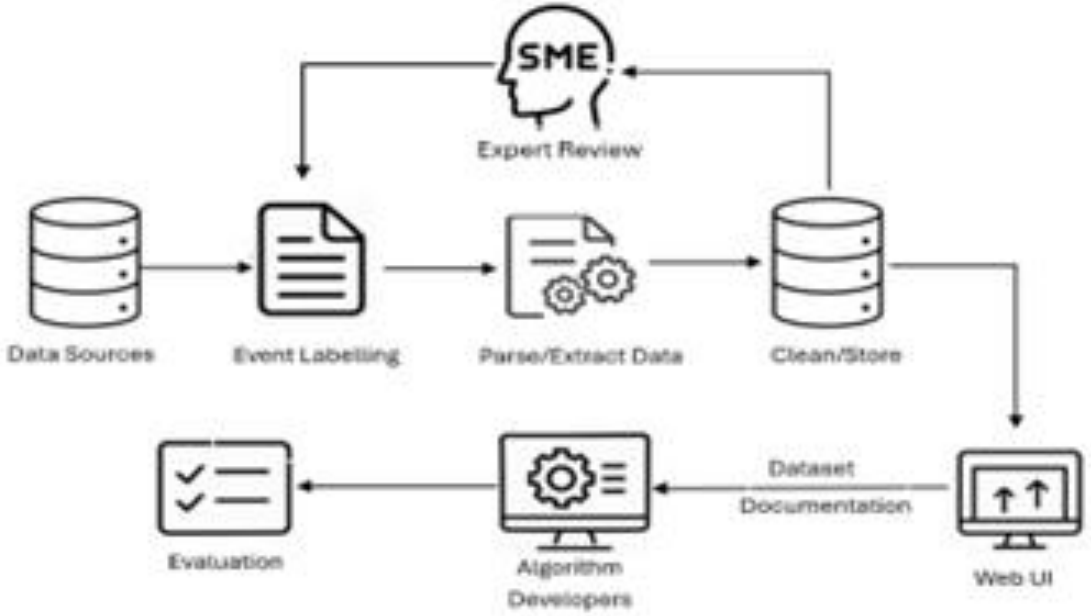
- The SDA TAP Lab, in partnership with the Space Operations Command (SpOC), is undertaking the development of a fully automated data annotation pipeline to facilitate the training and evaluation of machine learning algorithms for space domain awareness (SDA) tasks. This follows many industry best practices which are summarized in "50 Years of Data Science" as the Common Task Framework (Donoho, 2017).

- Currently, the open-source development community focused on the space domain is very small. There are no widely accepted ontologies for common tasks, labels, benchmarks, or performance metrics. As a result, we cannot say with any confidence what the state-of-the-art performance is for any algorithm employed in our enterprise. This is a challenge.

- The overall goal of the project is to accelerate the delivery of space battle management software to operational units by automating Common Tasks throughout the Kill Chain. Particular common tasks of high priority include Uncorrelated Track (UCT) Processing, State Estimation, and Object Identification; a more comprehensive list of tasks can be found at sdataplab.org. The Common Task Framework, as defined by Donoho's 2017 paper 50 Years of Data Science, provides a standardized method to evaluate and compare the performance of such automation algorithms.

Project Background

- A wealth of information related to these tasks exists on the Unified Data Library (UDL) and other online data sources, including observational data for both known objects and UCTs, current TLE and state vectors for tracked objects, launch events, maneuver events, etc. In order to use this data, software tools must be developed to streamline or automate the creation of benchmark datasets used for training and evaluation in the Common Task Framework. These software tools must be able to pull event data from available data sources, label that data according to predefined classifications (i.e. launch events, maneuver events, proximity events, etc.), parse and/or extract relevant measurement data depending on the classification, clean and store the labelled data in a centrally available database. As an iterative process over time, as subject matter experts refine data label definitions, higher fidelity data will become available. From this database, multiple benchmark datasets can be generated to either train or test various algorithms developed to automate common task workflows.
- The process of labelling and storing data (the goal of this project), as well as benchmark dataset generation and evaluation criteria (the goals of the sister project of Space Operations Command), must be well documented and repeatable so that it can be generalized and applied to other common tasks as they are identified. Additionally, documentation detailing the backend functionality of dataset generation and evaluation criteria must be made available to algorithm developers to enable a clear understanding of how algorithms can be improved.

Project Goal

- Develop Labelling & Data Storage approaches for processing Uncorrelated Tracks (UCTs) in the Common Task Framework. The end result of the Common Task Framework is a Web-hosted User Interface that algorithm developers can interact with to generate and download benchmark datasets related to the common task being solved in order to train their algorithms, as well as upload the results of the solution presented to be objectively evaluated and compared to other solutions presented to the same common task.